

AIM DMA Connector Specification

Ver0.1

DM270

Department: IMAGING

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HISTORY

Version	Date	Author	Notes
Ver0.0	03/12/02	Toshiki Suzuki	Initial Release for DM270
Ver0.1	05/14/02	Toshiki Suzuki	Spec change

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1. ABSTRACT

AIM DMA Connector(ADMAC)モジュールはMTCがAIMに直接アクセスすることを可能にする。

2. REFERENCES

ARM7TDMI datasheet ARM DDI 0029E, Aug. 1996, ARM Ltd.

3. GLOSSARY

AIM	ARM Internal Memory
ARM	ARM7TDMIE
BUSC	Bus Controller
HPIB	Host Peripheral Interface Bridge
MTC	Memory Traffic Controller
EHIF	External Host I/F

4. FUNCTION

- ◆ MTCのREQ信号によりAIMにアクセスし、動作完了後EN信号を発生
- ◆ AIMからのnWAIT信号に対応
- ◆ AIMのREVRに対してはアクセスしない
- ◆ MTCからのデータは32ビット固定長に限られるため、AIMへの8/16ビットアクセスはサポートしない

5. Interface

Table 1 : ADMAC Top-level I/O List

Pin Name	Bits	I/O	Fr/To	Description
Functional clocking / common signals				
rst_n	1	I	ext	Power on reset..
clk_arm	1	I	clkc	ARM clock.
Btsel	1	I	ext	Boot select
AIM interface signals				
aim_do_admac	32	I	ADMAC	Internal memory data input bus.
aim_nwait_admac	1	I	ADMAC	Wait signal from AIM. Active low.
admac_ma	11	O	ADMAC	Address bus
admac_mdo	32	O	ADMAC	Data output bus
admac_ncs_aim	6	O	ADMAC	Chip select signal, Active low
admac_nds	4	O	ADMAC	Data strobe, Active low
admac_mrnw	1	O	ADMAC	Read/write direction, 1:read / 0:write.
MTC interface signals				
mtc_dma_a	15	I	MTC	Address bus
mtc_dma_do	32	I	MTC	Data input bus
mtc_dma_rnw	1	I	MTC	Read/write direction, 1:read / 0:write,
mtc_dma_req	1	I	MTC	Chip select signal
admac_dma_do	32	O	MTC	data output bus.
admac_dma_en	1	O	MTC	DMA end flag to MTC
TEST signals				
tst_scan	1	I	tst	Scan mode

6. TEST Methodology

None.